

RegretSet-MMM: Learning to Select Marketing Mix Models by Their Downstream Economic Consequences

Evidence-adaptive set-wise learning, economic-regret supervision, and source-resolved attribution

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Abstract

Marketing mix modeling (MMM) is commonly selected by predictive fit, residual diagnostics, calibration agreement, or fixed analyst scores, although its operational purpose is budget decision support. RegretSet-MMM instead learns across advertisers which complete Bayesian MMM specification is least likely to produce costly budget decisions. Forty-eight truth-blind candidate MMMs per advertiser are fitted with PyMC FullRankADVI, a full-rank automatic-differentiation method in the stochastic variational inference family, and represented by 232 deployment-observable tokens covering diagnostics, posterior decision geometry, temporal identification, evidence attribution, and assumptions. Evidence-adaptive RegretSet-MMM retains the original joint all-token path and adds a business-level gate that routes residual representations of predictive, causal-identification, posterior-decision, and structural-specification evidence according to the evidence available for each advertiser. A permutation-invariant multi-head learner predicts loss across four predeclared budget decisions. We establish a finite-candidate bound connecting selected capped regret to regret-weighted oracle misrankings and decompose local ROI information among observational data, experiments, benchmarks, and regularization. The frozen evidence-adaptive selector was evaluated without retraining in a fresh cohort of 100 businesses generated by Google's independently developed Aggregate Marketing System Simulator. Across 4,800 candidate fits and 19,200 frozen actions, its capped decision-risk objective was 0.203, versus 0.353 for prediction-only selection, 0.401 for RegretSet-MMM, 0.458 for uniform random valid selection, 0.701 for classic Pareto selection, and 0.029 for an unattainable in-pool oracle. The two predeclared comparator tests and the P95 tail-safety check passed. These findings establish confirmatory external synthetic transport under the declared candidate and utility contracts, not real-advertiser effectiveness or universal state-of-the-art performance.

Keywords: marketing mix modeling; decision-focused learning; Bayesian inference; DeepSets; economic regret; source attribution; budget optimization; synthetic validation

1 Introduction

Marketing mix models estimate how marketing activity contributes to business outcomes and translate those estimates into resource-allocation decisions. Their scientific difficulty is familiar: spending is planned rather than randomized, channels co-move with latent demand, advertising has delayed and nonlinear effects, and external experiments or benchmarks are sparse and imperfectly transported. Consequently, several observationally plausible models can fit the same historical series yet imply substantially different returns on investment (ROI), marginal response curves, and optimal budgets [1–6].

Most model-selection procedures remain one step removed from the decision. They favor out-of-sample prediction, likelihood-based criteria, calibration distance, structural diagnostics, or a fixed weighted score. These quantities are useful evidence, but none is identical to the economic opportunity lost when a selected model reallocates budget incorrectly. Nor is there a universal reason that a diagnostic should have the same importance for a low-signal search advertiser, a heavily flighted television advertiser, and a business with a strong independent experiment.

RegretSet-MMM treats model selection as a supervised decision problem across advertisers. In a synthetic world, causal response surfaces are known. Candidate MMMs are fitted without access to that truth, asked to choose budgets under declared scenarios, and then labeled by the value forgone relative to the best feasible action under the hidden response surface. A set-wise learner maps only deployment-observable diagnostics and posterior summaries to the distribution of this loss. For a new advertiser, the learned rule ranks candidate specifications without observing causal truth.

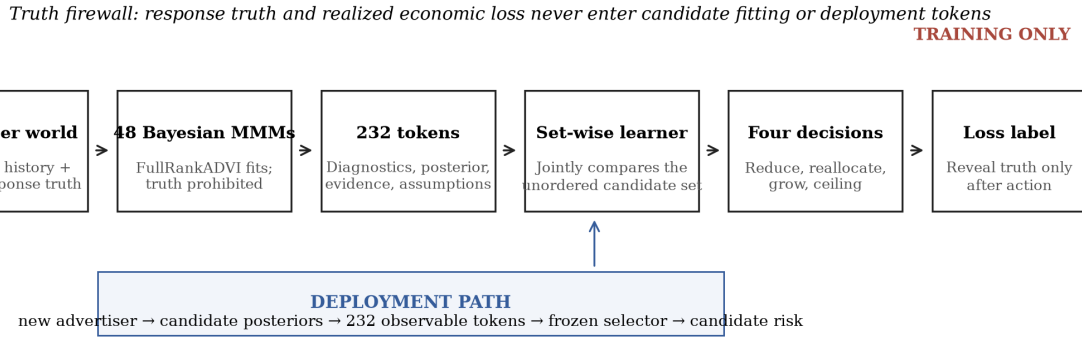


Figure 1. Research and deployment paths. Hidden causal truth is opened only after a candidate has acted in simulation. Deployment uses the frozen mapping from observable candidate sets to predicted decision risk.

1.1 Contributions

- A decision-focused formulation of cross-advertiser MMM selection in which the supervised target is downstream economic loss rather than a hand-weighted model-quality score.
- An evidence-adaptive, permutation-invariant learner that evaluates all 48 candidates jointly and emits location, tail, scenario, and dangerous-false-champion predictions for every candidate.
- A 232-token, truth-blind representation combining classical diagnostics with FullRankADVI posterior geometry, channel-response uncertainty, cross-channel posterior dependence, evidence provenance, and model assumptions.
- A finite-candidate regret bound and a theorem-aligned, gap-weighted ranking surrogate that penalize economically costly misrankings more strongly than harmless ones.
- A continuous local decomposition of ROI information into observational data, experiments, industry benchmarks, and regularization, together with separate notions of evidence quality, conflict, and decision dependence.
- An auditable validation design using advertiser-grouped nested cross-validation, complete-mechanism holdouts, a one-time internal sealed cohort, and a fresh confirmatory external transport audit in Google's independently developed Aggregate Marketing System Simulator.

1.2 Claim boundary

The contribution is a method for learning and auditing an MMM selection rule. An internal sealed audit evaluates RegretSet-MMM on 140 previously unseen advertisers generated by the study's synthetic data-generating process. A separate external audit evaluates evidence-adaptive RegretSet-MMM in the independently developed AMSS simulator against RegretSet-MMM, prediction-only selection, classic Pareto selection, and uniform random valid selection, with an in-pool oracle reported only as an unattainable lower reference. These synthetic results do not imply real-advertiser effectiveness, universal state-of-the-art performance, inclusion of the true model in the candidate library, or exact posterior representation by FullRankADVI.

2 Related work

2.1 Bayesian marketing mix modeling

Modern Bayesian MMMs combine baseline controls, seasonality, adstock, saturation, and constrained or regularized media effects [1,2]. Open implementations have expanded access to geometric and Weibull carryover, response curves, time-varying coefficients, experiment calibration, and posterior budget optimization [4–8]. Recent work emphasizes that experiments should calibrate compatible estimands and time windows, while paid-search and other demand-harvesting channels remain especially vulnerable to endogeneity [3,5,9]. RegretSet-MMM does not replace those estimators; it learns which complete specification is least risky for a declared decision.

2.2 Model assessment and causal validation

Predictive validation detects instability and distribution shift but need not recover a causal decomposition. Residual tests diagnose misspecification but cannot establish exogeneity from observed residuals alone. Experiment agreement tests external coherence, although experiment overlap, low power, and poor transport may mislead. Information criteria and cross-validation estimate predictive generalization [10]; they do not directly score the counterfactual value of a budget allocation. The proposed token registry retains these signals while allowing their relevance to be learned from economic labels.

2.3 Decision-focused learning and learning to rank

Predict-then-optimize methods train predictive systems using downstream optimization loss rather than parameter error alone [11,12]. Ranking methods similarly exploit relative ordering within structured candidate sets [13]. RegretSet-MMM adapts these ideas to causal model selection: the objects being ranked are fitted Bayesian MMMs, each candidate generates four economic actions, and the training label is excess decision loss under a hidden causal response surface.

2.4 Set representation and variational posterior inference

DeepSets provides a general architecture for permutation-invariant functions over unordered collections [14]. Here candidate order is arbitrary, so a shared encoder and symmetric pooling operator supply advertiser-specific context without creating a candidate-ID signal. Candidate posteriors are fitted with PyMC FullRankADVI, a full-rank automatic-differentiation variational inference procedure within the broader stochastic variational inference family. It scales to tens of thousands of fits but may understate tails or miss multimodality [15,16]. The selection framework is inference-engine agnostic in principle: the same posterior tokens can be computed from more accurate draws when resources permit.

2.5 Positioning of the contribution

The scientific question is whether deployment-observable evidence predicts the economic consequences of selecting among plausible MMMs for unseen advertisers. The internal audit compares RegretSet-MMM with a frozen restricted-diagnostic baseline that uses the same candidate set and economic objective but only a smaller predeclared subset of observable diagnostics. This comparison tests whether the complete token representation adds economic-ranking information; the restricted baseline is not proposed as a production selector.

Ground-truth simulation has an established role in marketing measurement research. AMSS generates aggregate marketing systems with known ROI and marginal ROI [23], while a recent independently released benchmark explicitly models endogenous budget feedback, promotional anticipation, scheduled television flights, and performance chasing [24]. These systems are especially relevant external challenges because their equations and implementation were not selected to favor RegretSet-MMM.

3 Decision problem

3.1 Advertisers, candidates, and actions

Let b index advertisers and let S_b be a finite set of C candidate MMMs fitted to advertiser b . In the implemented design $C = 48$. Candidate c observes history D_b , declared external evidence E_b , and the business decision contract. It produces a posterior over parameters and a recommended allocation a_{bcq} for each scenario q . Synthetic causal truth θ_b is inaccessible until every candidate action is fixed.

$$\begin{aligned} p_c(\theta \mid D_b, E_b), \quad a_{bcq} &= \mathcal{A}_q[p_c(\theta \mid D_b, E_b)] \\ \theta_b^{\text{true}} &\notin \{D_b, E_b, x_{bc}\} \end{aligned} \tag{1}$$

3.2 Economic value and loss

For action a , let $V_{bq}(a; \theta_b)$ denote its true incremental economic value in scenario q . The value function applies each allocation to the simulator's channel-specific response surfaces and converts incremental outcome to contribution using the advertiser's effective revenue margin before subtracting media cost. The feasible oracle action a^*_{bq} is the highest-value action in the same declared allocation grid and constraints seen by all candidates.

$$\begin{aligned} a^*_{bq} &= \arg \max_{a \in \mathcal{A}_{bq}} V_{bq}(a; \theta_b^{\text{true}}) \\ \ell_{bcq} &= V_{bq}(a^*_{bq}; \theta_b^{\text{true}}) - V_{bq}(a_{bcq}; \theta_b^{\text{true}}) \geq 0 \end{aligned} \tag{2}$$

The candidate-level label aggregates four predeclared decisions: reducing total spend, reallocating a fixed budget, growing spend, and finding an economic ceiling. The implementation retains all four scenario losses rather than training only a single average. A nonnegative weighted aggregate L_{bc} defines the principal decision loss, while scenario heads expose where the risk originates.

$$L_{bc} = \sum_{q=1}^4 \omega_q \ell_{bcq}, \quad \omega_q \geq 0, \quad \sum_q \omega_q = 1 \quad (3)$$

3.3 Excess loss and deployment risk

The in-pool oracle c_b^* is the candidate with minimum loss among eligible candidates, not the true structural model. Candidate excess loss Δ_{bc} measures the opportunity cost of selecting c instead of that oracle. A business-specific, candidate-independent scale s_b converts losses to a capped normalized target used by the ranking theorem and training regularizer. Uncapped monetary and normalized losses remain reportable safety endpoints.

$$\begin{aligned} c_b^* &= \arg \min_{c \in S_b} L_{bc}, & \Delta_{bc} &= L_{bc} - L_{bc^*} \\ \bar{\Delta}_{bc} &= \min\left(\frac{\Delta_{bc}}{s_b}, 1\right) \end{aligned} \quad (4)$$

3.4 Declared risk preference

The selector predicts both the conditional mean loss and the P90 loss. The default business protection rule assigns 65% weight to the mean and 35% to P90. This is a declared utility preference, not a learned scientific constant. A more risk-averse organization may increase the tail weight; a risk-neutral organization may reduce it. Sensitivity to this mixture must be reported without selecting the weight on sealed-audit performance.

$$R_{bc} = 0.65 \widehat{\mathbb{E}}[L_{bc} | X_b] + 0.35 \widehat{Q}_{0.90}[L_{bc} | X_b] \quad (5)$$

4 Regret theory for finite candidate selection

Assumption 1 (finite eligible set). For each advertiser b , the eligible candidate set S_b is nonempty and finite; ties are resolved by a predeclared deterministic rule.

Assumption 2 (candidate-independent scale). The positive economic scale s_b may depend on the advertiser and decision contract but not on which candidate is evaluated.

Definition 1 (selector). Let f_{bc} be a scalar predicted risk for candidate c , with lower values preferred, and let the selected candidate be $\hat{c}_b = \arg \min_{c \in S_b} f_{bc}$.

4.1 Selected-regret bound

The following inequality motivates an all-pairs, regret-weighted ranking loss. It is deliberately modest: it connects a finite candidate selector to the oracle comparisons it gets wrong; it does not prove out-of-population generalization.

Proposition 1 (finite-candidate misranking bound). For every advertiser satisfying Assumptions 1–2,

$$\begin{aligned} \bar{\Delta}_{b\hat{c}_b} &\leq \\ \sum_{c \in S_b \setminus \{c_b^*\}} \bar{\Delta}_{bc} \mathbb{I}[f_{bc} \leq f_{bc^*}] &\end{aligned} \quad (6)$$

Proof. If the selector chooses the oracle, the left side is zero and the inequality is immediate. Otherwise the selected candidate $\hat{c}_b$ is a non-oracle candidate satisfying $f_{b\hat{c}_b} \leq f_{bc^*}$. Its term therefore appears in the sum with weight equal to the left side. Every remaining term is nonnegative. $\square$

The result says that a costly selected mistake is necessarily covered by the total weight of non-oracle candidates that the score ranks at least as favorably as the oracle. It does not require all ordering errors to be equally costly: a near-oracle swap receives a small gap, whereas an economically destructive false champion receives a large gap.

4.2 Differentiable surrogate

For temperature $T > 0$ and $u = (f_{bc^*} - f_{bc})/T$, the event that c outranks the oracle implies $u \geq 0$. Since $\text{softplus}(u)$ is at least $\log 2$ whenever $u \geq 0$, the indicator is upper-bounded by $\text{softplus}(u)/\log 2$.

$$\begin{aligned} \mathbb{I}[u \geq 0] &\leq \frac{\log(1 + e^u)}{\log 2} \\ \widetilde{\Delta}_{b\hat{c}_b} &\leq \sum_{c \neq c_b^*} \widetilde{\Delta}_{bc} \frac{\text{softplus}((f_{bc_b^*} - f_{bc})/T)}{\log 2}. \end{aligned} \quad (7)$$

Training therefore includes every oracle-versus-candidate comparison with its exact predeclared normalized gap. The implementation additionally optimizes a direct set-wise objective: a softmax distribution over negative predicted risk is charged the expected capped normalized regret of selecting from the complete candidate set. The theorem-aligned pair loss is a regularizer; the direct selection loss is the closer approximation to deployment.

4.3 What the theorem does and does not prove

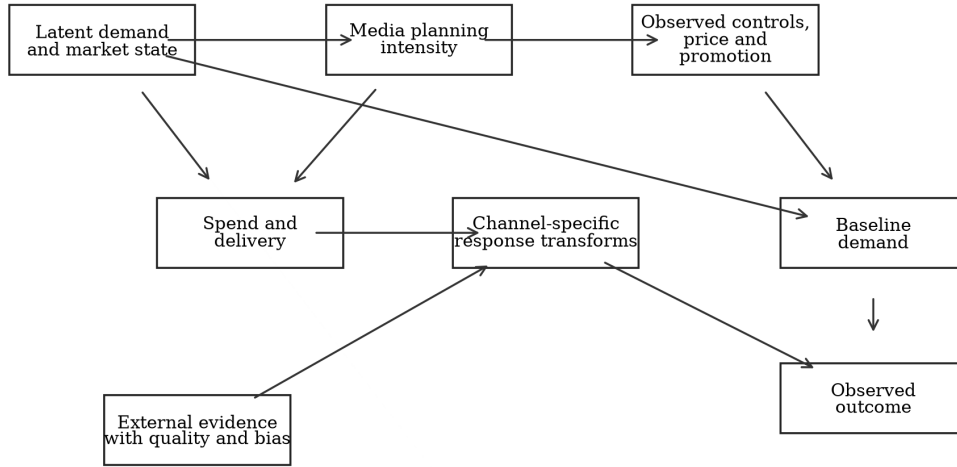
- It is deterministic and applies to the realized finite pool, regardless of how the score was trained.
- It justifies gap-weighted oracle comparisons and supplies a testable receipt: selected capped regret must not exceed the reported hard or surrogate bound.
- It does not show that S_b contains the true response model, that f generalizes, or that capped loss protects uncapped catastrophic tails; P90, P95, CVaR, and maximum uncapped loss remain necessary.

5 Synthetic advertiser population

5.1 Data-generating process

A synthetic benchmark is valuable because causal contributions and counterfactual budget value are known. It is also dangerous because a selector can learn artifacts of one generator. The population therefore varies business duration, seasonality, demand persistence, media planning, commercial shocks, channel delivery, response curves, controls, noise, external evidence, and decision economics. The generator is separate from the candidate transformation code.

Endogeneity arises because latent demand and planning affect both media and outcome



Hidden truth includes the channel response surface, causal contribution and counterfactual value of each allocation.

Figure 2. Causal structure of the advertiser generator. Spend is endogenous because latent demand and planning affect both media delivery and outcome. External evidence is generated separately and may be noisy, biased, or poorly transported.

5.2 Observed outcome

For advertiser b at time t , observed outcome is the sum of baseline demand, control effects, channel contributions, and stochastic disturbance. Each channel contribution applies a delivery process, a channel-specific carryover kernel, and a Hill-type saturation function. Channel parameters are not forced to be common: television can have delayed Weibull carryover while paid search can have short geometric memory, and saturation levels differ by channel and advertiser.

$$\begin{aligned} Y_{bt} &= B_{bt} + \gamma_b^\top Z_{bt} + \sum_{j=1}^{J_b} m_{bjt} + \varepsilon_{bt} \\ m_{bjt} &= \beta_{bj} h(w_{bj} * X_{bj})_t; \kappa_{bj}, \eta_{bj} \end{aligned} \quad (8)$$

The histories span 104–260 weeks. Demand, planning, and commercial-state persistence vary broadly; event shocks, promotions, observed demand proxies, measurement noise, gross margin, lifetime-value multipliers, and maximum budget changes are heterogeneous. The principal implementation contains three archetypal channels—paid social, paid

search, and television—because their delivery and endogeneity mechanisms differ sharply. This simplification is a limitation, not a claim that real MMMs contain only three channels.

5.3 Seven mechanism families

Family	Primary stress	Why it matters
Balanced DTC	moderate persistence and signal	reference business with mixed digital and offline media
Search demand harvesting	search follows latent demand	tests endogeneity and inflated observational search ROI
Social frequency pressure	reach/frequency saturation	tests nonlinear marginal returns
Delayed television	long, delayed carryover	tests temporal support and post-flight attribution
Correlated planning	shared budget and promotion process	tests multicollinearity and conditional identification
Wrong external evidence	biased or transported anchors	tests conflict, quality, and source dependence
Swapped mechanics	unexpected channel response families	tests transfer beyond channel-name stereotypes

Table 1. Predeclared simulator families. Each family changes a causal mechanism rather than merely a random seed.

5.4 Carryover and saturation ranges

Carryover is generated by either normalized geometric or Weibull convolution. Geometric decay ranges from approximately 0.01 to 0.78, with the upper tail concentrated in television. Weibull shape spans approximately 1.15–4.80; scale spans 1.5–8 weeks for digital channels and 3.5–13 weeks for television. Hill shapes range approximately 0.90–3.10 for paid social and 0.72–2.35 for other channels. Half-saturation points are tied to advertiser-specific spend quantiles between roughly 0.30 and 0.82. These ranges create short memory, delayed peaks, weak saturation, and frequency pressure without assuming one channel response globally.

5.5 External evidence

Experiments are available probabilistically and represent noisy measurements of ROI over their tested spend and outcome window, including a declared post-test carryover period when appropriate. They have heterogeneous power, bias, replication, platform or geo design, and transportability. Benchmarks are noisy channel-level priors; a dedicated wrong-evidence family shifts them downward or upward. Candidate evidence arms determine whether available qualified evidence is used, but hidden truth never chooses an arm.

$$ROI_{bjW}^{\text{exp}} = ROI_{bjW}^{\text{true}}(1 + b_{bjW}) + \epsilon_{bjW} \quad (9)$$

$$ROI_{bjW}^{\text{true}} = \frac{\sum_{t \in W} m_{bjt}^{\text{incremental}}}{\sum_{t \in W} X_{bjt}^{\text{tested}}}$$

5.6 Decision environments

The allocation engine evaluates 21 budget levels and 240 candidate allocations per level under channel-specific cut and growth constraints. Economic value is incremental outcome multiplied by an effective revenue margin minus incremental media cost. The four scenarios cover budget reduction (up to 35%), fixed-budget reallocation, budget growth (up to 50%), and an economic-ceiling search (up to 80% growth). This makes the label depend on marginal response and channel substitution, not only average ROI recovery.

6 Candidate Bayesian MMMs and posterior inference

6.1 Candidate library

Each advertiser receives the same 48-candidate opportunity set: 24 model specifications crossed with two predeclared evidence settings. Specifications vary adstock family and parameters, Hill saturation, baseline Fourier order and cycle, ridge strength, likelihood family, prior family, optional time-varying coefficients, planning-intensity controls, and calibration route. The library is deliberately misspecified for some worlds; otherwise model selection would be trivial.

Component	Candidate variation	Scientific ambiguity represented
Carryover	geometric or Weibull; short to long memory	immediate decay versus delayed peak and long tail
Saturation	multiple Hill shapes and half-saturation values	weak versus strong diminishing returns
Baseline	Fourier order, cycle, controls	seasonality and recurring demand allocation
Coefficients	constant or time-varying	stable effect versus evolving response
Planning	latent-intensity factor on/off	shared demand and media-planning process

Component	Candidate variation	Scientific ambiguity represented
Distributions	positive priors; Gaussian, Student-t, or log-normal outcomes	tail behavior, skew, and robustness
Evidence	qualified experiment or benchmark use	external identification and transport

Table 2. Major candidate dimensions. Exact grids and fingerprints are part of the open research contract.

6.2 Full-rank variational inference

All research candidates use PyMC 6.2 FullRankADVI under a frozen contract. A multivariate Gaussian variational family $q_\phi(\theta)$ is optimized by maximizing the evidence lower bound (ELBO), retaining covariance that a mean-field approximation would discard. Posterior draws from q_ϕ feed response, uncertainty, and decision tokens. Multiple convergence and reproducibility checks guard against unstable envelopes.

$$\phi^* = \arg \max_{\phi} \{ \mathbb{E}_{q_\phi(\theta)} [\log p(D, E, \theta)] - \mathbb{E}_{q_\phi(\theta)} [\log q_\phi(\theta)] \} \quad (10)$$

FullRankADVI is used because tens of thousands of posterior fits are required. It is not assumed superior to Markov chain Monte Carlo. Like other variational methods, FullRankADVI can smooth multimodal posteriors and distort tail mass, so the inference engine is part of the reported research contract. The learned selector is compatible with posterior draws from NUTS in principle, provided the 232-token contract is computed identically.

6.3 Truth firewall

Business identity, generator family, split labels, candidate identity, search metadata, synthetic parameters, decision loss, ROI truth error, and contribution truth error are forbidden inputs. Tests permute candidate and channel order, scan token names, and verify that model predictions are unchanged by those permutations. The only path from hidden truth to the learner is through training labels after actions have been fixed.

7 Deployment-observable token representation

7.1 Tokens rather than opaque embeddings

A token is a named numerical input with a declared construction and interpretation. The term emphasizes that the model receives a structured vocabulary of evidence rather than raw outcome and spend histories. Tokens may be individual diagnostics, response assumptions, posterior summaries, or predeclared interactions. The learner creates hidden embeddings from these tokens, but every input remains auditable and reproducible.

The representation contains 136 base tokens and 96 posterior decision tokens, for 232 per candidate. Base tokens cover predictive generalization, residual structure, causal and placebo stress, ROI coherence, temporal evidence, evidence quality and conflict, model assumptions, and their allowed interactions. Posterior tokens summarize channel-level ROI geometry, response uncertainty, evidence provenance, contribution stability, and cross-channel dependence from FullRankADVI draws.

Token pillar	Examples	Question answered
Predictive generalization	rolling holdout error; high/low-spend regimes	does the candidate forecast outside its fitting regime?
Structural adequacy	residual pattern; variance; collinearity; rank	does the fitted structure leave diagnostic failures?
Causal and temporal robustness	future-media placebo; anchor recovery; carryover support	are timing and causal claims compatible with available evidence?
Decision coherence	ROI plausibility; marginal-return and source-deletion changes	does the posterior support stable, plausible allocations?
Posterior decision geometry	ROI tail mass; interval overlap; contribution CV; correlations	where are location, tail, and substitution risks concentrated?
Specification context	adstock, saturation, likelihood, dynamics, calibration	which assumptions generated those diagnostics?

Table 3. Practitioner-facing token taxonomy. Pillars organize interpretation only; they are not fixed score weights.

7.2 Channel-order invariance

Twenty-three channel metrics are reduced by four symmetric operators—mean, standard deviation, minimum, and maximum—yielding 92 tokens. Four additional summaries measure mean and maximum absolute posterior correlation among channel log-ROI and contribution draws. The representation is invariant to channel order and does not require the same number of channels.

$$T_{mk}(c) = \text{Agg}_{j \in \mathcal{J}_b} t_m(q_c(\theta_j)), \quad \text{Agg} \in \{\text{mean, sd, min, max}\} \quad (11)$$

$$92 = 23 \times 4, \quad 96 = 92 + 4 \text{ pairwise dependence summaries}$$

7.3 Posterior token content

Posterior family	Representative quantities
ROI location and interval	log median, mean, lower and upper interval endpoints, relative width, draw SD
Decision safety	implausible probability, near-zero probability, skewness, analytic-to-FullRankADVI location shift, interval overlap
Response mechanics	adstock moments, Weibull scale and shape, saturation moments, half-saturation, response-family shares
Contribution reliability	log contribution coefficient of variation
Evidence provenance	experiment and benchmark source indicators
Substitution geometry	absolute pairwise log-ROI and contribution correlation summaries

Table 4. Ninety-six FullRankADVI posterior decision tokens. All are computed without synthetic truth.

7.4 Why interactions matter

A diagnostic rarely has one global meaning. Wide ROI intervals may be acceptable for a small channel but dangerous near a large reallocation threshold. Residual drift may matter differently under long-memory television than under short-memory search. The base registry therefore includes only predeclared interactions between interpretable diagnostics and model assumptions; the nonlinear set learner can additionally form bounded interactions after fold-specific standardization.

8 RegretSet-MMM learner

8.1 Permutation-invariant candidate-set architecture

Candidates are not independent rows: whether a model is attractive depends on the alternatives available for the same advertiser. RegretSet-MMM therefore consumes the entire unordered set $X_b = \{x_{bc}\}$. A shared encoder ϕ maps each 232-token vector to a candidate embedding. Mean pooling creates advertiser context h_b . The decoder receives the candidate, context, difference, and elementwise interaction embeddings, preserving permutation-equivariant predictions.

$$z_{bc} = \phi(x_{bc}), \quad h_b = |S_b|^{-1} \sum_{c \in S_b} z_{bc} \quad (12)$$

$$u_{bc} = \psi(z_{bc}, h_b, z_{bc} - h_b, z_{bc} \odot h_b)$$

Permutation-invariant advertiser context; permutation-equivariant candidate predictions

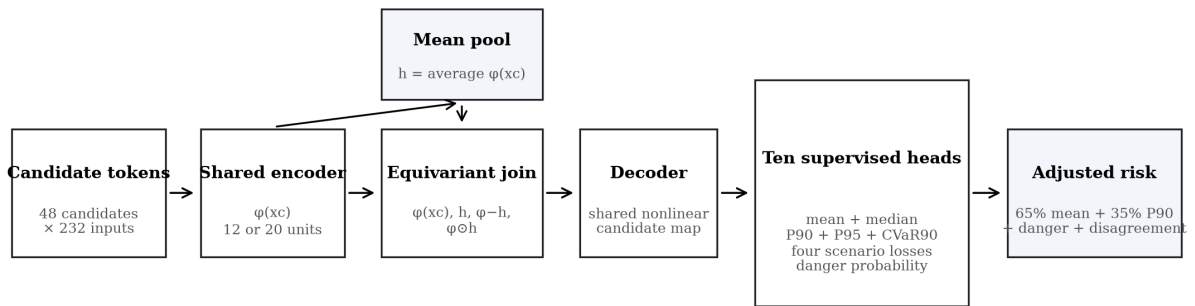


Figure 3. Core set-wise, multi-head pathway used by RegretSet-MMM and retained as the joint all-token path in evidence-adaptive RegretSet-MMM. Reordering candidates or channels cannot change candidate predictions. Equation (16a) adds the evidence-gated residual pathway without removing this joint representation.

8.2 Ten prediction heads

The decoder produces ten outputs per candidate: mean, median, P90, P95, and CVaR90 across the candidate's four predeclared scenario-loss values; the four scenario-specific absolute losses; and the probability of being a dangerous false champion. These summaries describe the declared scenario distribution, not posterior quantiles of an unknown causal truth. Quantile outputs use positive softplus increments so that $\text{median} \leq \text{P90} \leq \text{P95} \leq \text{CVaR90}$ by construction. A candidate is dangerous when it appears sufficiently attractive to win while its true normalized excess loss exceeds the predeclared threshold.

8.3 Supervised objective

The objective combines robust location loss, quantile loss, tail expectation, scenario decomposition, dangerous-champion classification, direct set-wise selection regret, and the theorem-aligned oracle-pair regularizer. Let y_{bc} denote mean loss, $q_{\tau, bc}$ the empirical quantile of the four scenario losses, d_{bc} the danger indicator, and p_{bc} the model-implied soft selection probability.

$$\begin{aligned} \mathcal{L}_{heads} = & \mathcal{L}_{Huber}(\hat{\mu}, y) + 0.15 \mathcal{L}_{Huber}(\hat{q}_{.5}, q_{.5}) \\ & + 0.35 \rho_{.90}(q_{.90} - \hat{q}_{.90}) + 0.20 \rho_{.95}(q_{.95} - \hat{q}_{.95}) \\ & + 0.30 \mathcal{L}_{Huber}(\widehat{CVaR}_{.90}, CVaR_{.90}) + 0.12 \sum_q \mathcal{L}_{Huber}(\hat{\ell}_q, \ell_q) + 0.45 BCE(\hat{d}, d), \end{aligned} \quad (13)$$

$$\begin{aligned} p_{bc} = & \frac{\exp(-f_{bc}/T)}{\sum_{r \in S_b} \exp(-f_{br}/T)}, \quad \mathcal{L}_{set} = \sum_b \sum_{c \in S_b} p_{bc} \tilde{\Delta}_{bc} \\ \mathcal{L}_{pair} = & \sum_b \sum_{c \neq c_b^*} \tilde{\Delta}_{bc} \text{softplus}(f_{bc_b^*} - f_{bc}), \end{aligned} \quad (14)$$

$$\mathcal{L} = \mathcal{L}_{heads} + \mathcal{L}_{set} + 0.25 \mathcal{L}_{pair} + \lambda \|\Theta\|_2^2 \quad (15)$$

The numerical coefficients in the head loss are frozen optimization choices, not claims about the scientific importance of the corresponding diagnostic pillars. They balance gradients on distinct supervised targets. Candidate selection at deployment follows equation (16b), whose 65/35 mean-tail mixture is explicitly a business risk preference.

8.4 From RegretSet-MMM to evidence-adaptive RegretSet-MMM

RegretSet-MMM uses one pooled 232-token DeepSets pathway for every advertiser. It can learn interactions involving experiment and benchmark indicators, but it has no separate representation of the advertiser's overall evidence environment. Evidence-adaptive RegretSet-MMM preserves that joint all-token pathway and adds four disjoint residual encoders for predictive generalization, causal identification, posterior decision quality, and structural specification. A 63-token, candidate-independent context records evidence availability and channel coverage, explicit missingness masks, candidate-set regime, set-level diagnostic summaries, and matched benchmark-minus-experiment summaries.

$$\begin{aligned} g_b = & \text{softmax}(W_g C_b + b_g), \quad \sum_{k=1}^4 g_{bk} = 1 \\ r_{bc} = & \sum_{k=1}^4 g_{bk} \phi_k(x_{bc}^{(k)}), \quad \tilde{z}_{bc} = [\phi_{all}(x_{bc}), r_{bc}], \end{aligned} \quad (16a)$$

The joint path prevents the expert partition from discarding cross-pillar interactions; the gated path permits the marginal value of those signals to vary with the available evidence. The softmax values are residual routing weights, not feature importance, source validity, or causal identification shares. The ten supervised heads, theorem-aligned loss, 65/35 mean-tail decision utility, and truth firewall remain unchanged.

Thus, the two selectors share the same fitted candidate posteriors, 232 candidate tokens, economic-loss targets, and set-wise learning principle. Their sole architectural distinction is that evidence-adaptive RegretSet-MMM also observes 63 advertiser-level context tokens and uses them to vary the residual contribution of four evidence-specific encoders.

Training reuses the 20,160 FullRankADVI candidates from 420 development advertisers without posterior refitting. Each advertiser contributes three matched candidate sets: all 48 candidates, the 24 experiments-only candidates, and the same 24 structural specifications with benchmark gap-fill. All three remain in the same advertiser fold, share the observed outcome, spend paths, hidden truth, and simulator family, and recompute the in-regime oracle and excess loss. This paired construction teaches the gate about benchmark availability without confounding the comparison with a different advertiser. It does not identify the counterfactual effect of removing an existing experiment, because doing so would require refitting the posterior; post-fit token masking was prohibited.

Across cross-fitted full candidate sets, the mean residual routing weights were 22.3% predictive generalization, 21.1% causal identification, 28.8% posterior decision quality, and 27.8% structural specification. These averages describe how the auxiliary residual path was routed in development. They neither sum with the joint path nor replace the source-resolved posterior attribution in Section 9.

8.5 Ensemble policy and abstention

Five bootstrap selector fits estimate model disagreement. Predicted danger and disagreement increase adjusted risk. Confidence combines predicted danger, relative ensemble uncertainty, and ambiguity between the best and second-best candidates. Selective promotion can abstain on low-confidence advertisers; all primary results nevertheless report 100% coverage to prevent apparent improvement through selective omission.

$$\begin{aligned} f_{bc} &= R_{bc} + \lambda_d \hat{p}(d_{bc} = 1) + 0.25 \text{SD}_m(R_{bc}^{(m)}) \\ \text{conf}_b &= 1 - (0.4 \hat{p}_d + 0.3 u_b^{\text{rel}} + 0.3 a_b), \end{aligned} \quad (16b)$$

8.6 Nested model and policy selection

Two compact architectures and three danger penalties are compared only inside four-fold inner grouped cross-validation. Five outer grouped folds estimate transfer to unseen advertisers. All rows from one advertiser remain in one fold. Fold-specific standardization is fitted only on the training advertisers, and cross-fitted predictions are retained for every development business.

9 Source-resolved evidence attribution

9.1 Posterior information decomposition

A channel estimate may be precise because the observational design identifies it, because an experiment anchors it, because an industry benchmark supplies prior information, or because generic regularization shrinks it. RegretSet-MMM reports these sources continuously. At the local Laplace approximation, posterior curvature is decomposed into positive semidefinite components for observational data D, experiments E, benchmarks B, and regularization R.

$$H_{\text{post}} = H_D + H_E + H_B + H_R, \quad \Sigma = H_{\text{post}}^{-1} \quad (17)$$

For channel-j ROI gradient g_j , the implemented variance attribution propagates each curvature source through the common posterior covariance. The raw components sum to the local posterior ROI variance up to numerical error; normalization yields nonnegative source shares.

$$\begin{aligned} v_{s,j} &= g_j^\top \Sigma H_s \Sigma g_j, & v_j &= g_j^\top \Sigma g_j \\ a_{s,j} &= \frac{v_{s,j}}{\sum_{r \in \{D, E, B, R\}} v_{r,j}}, & a_{s,j} &\geq 0, \quad \sum_s a_{s,j} = 1. \end{aligned} \quad (18)$$

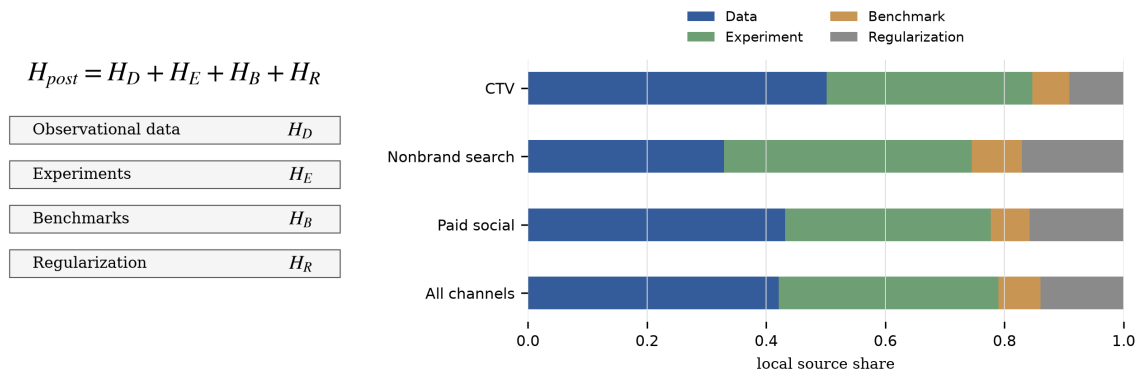


Figure 4. Observed local evidence attribution for the 1,260 channel posteriors in the 420 outer-fold selected development candidates. Bars report equal-channel mean Laplace curvature shares. Overall, observational data supplied 42.1%, experiments 36.9%, benchmarks 7.0%, and regularization 14.0% of local ROI information. These are development-cohort influence estimates, not scores of source validity.

The empirical decomposition shows that the selected candidates were neither simply data-led nor prior-led. Observational information supplied the largest average share, but experiments supplied more than one third of local ROI information and were the largest component for nonbrand search. CTV had the strongest observational share. Benchmark information was deliberately narrower because it entered only as gap-filling evidence when qualified experiments were absent; weak

regularization supplied the remaining local curvature. Spend weighting produced a similar split—45.0% data, 34.0% experiments, 6.1% benchmarks, and 14.9% regularization—so the result is not an artifact of treating small and large channels equally.

Conditional observational identification was far lower than the raw data share: the mean residualized data share was 8.5%, with a 3.9% median. The distinction is substantive. The data can contribute posterior precision jointly while offering little channel-specific variation after the other regressors are projected out. This is precisely why source attribution and conditional identification are reported separately.

9.2 Conditional observational identification

Raw column magnitude overstates identification when channels are correlated. The implementation projects a channel's normalized ROI direction on every other design column and measures the weighted residual information. This is a Schur-complement interpretation: only variation that cannot be reproduced by the remaining regressors receives conditional observational credit. Perfectly duplicated channels have no separate likelihood identification even if both columns are large.

Proposition 2 (duplicated-channel non-identification). If two transformed media columns satisfy $X_1 = X_2$, the likelihood depends on β_1 and β_2 only through $\beta_1 + \beta_2$. Individual channel effects are unidentified without external information or additional restrictions.

Proof. Substituting $X_2 = X_1$ gives $X_1\beta_1 + X_2\beta_2 = X_1(\beta_1 + \beta_2)$. Every coefficient pair with the same sum has the same likelihood. $\square$

9.3 Local source sensitivity

In a scalar conjugate Gaussian example, posterior mean is a precision-weighted average of source means. The derivative with respect to source s is exactly its normalized precision. This supplies a simple interpretation of local evidence share as sensitivity, while equation (18) extends the idea to correlated multivariate coefficients and an ROI functional.

$$\begin{aligned}\mu_{post} &= \frac{\sum_s \tau_s \mu_s}{\sum_s \tau_s} \\ \frac{\partial \mu_{post}}{\partial \mu_s} &= \frac{\tau_s}{\sum_r \tau_r}.\end{aligned}\tag{19}$$

9.4 Bias transmission, conflict, and decision dependence

Influence is not validity. If source s has bias δ_s in the conjugate setting, posterior bias is the influence-weighted sum of source biases. RegretSet-MMM therefore keeps source influence distinct from source quality, standardized conflict, and the downstream decision change when a source is removed.

$$\begin{aligned}\text{Bias}(\mu_{post}) &= \sum_s \frac{\tau_s}{\sum_r \tau_r} \delta_s \\ C_{s,j} &= \frac{|\mu_{D,j} - \mu_{s,j}|}{\sqrt{\sigma_{D,j}^2 + \sigma_{s,j}^2}} \\ D_s &= \|\mathcal{A}[p(\theta | D, E, B)] - \mathcal{A}[p(\theta | D, E_{-s}, B_{-s})]\|.\end{aligned}\tag{20}$$

Experiment quality records independence, MMM-data overlap, test power, tested spend range, campaign and outcome windows, post-test carryover, and temporal or audience transportability. Benchmark quality records channel, tactic, objective, and business-model match, recency, sample size, uncertainty, and whether the source is specific or a broad fallback. Descriptive phrases such as data-led or experiment-led may summarize equation (18), but they receive no hand-designed validation points and are never interpreted as evidence validity.

10 Validation and governance

10.1 Development cohort

The development dataset contains 420 advertiser worlds and 20,160 posterior-fitted candidates. Advertiser-grouped nested cross-validation separates hyperparameter selection from outer-fold assessment. A complete leave-one-generator-family-out challenge asks whether the learner transfers to a mechanism absent from its training families. Neither procedure is a substitute for a sealed audit, because researchers can still make method-level choices after seeing development summaries.

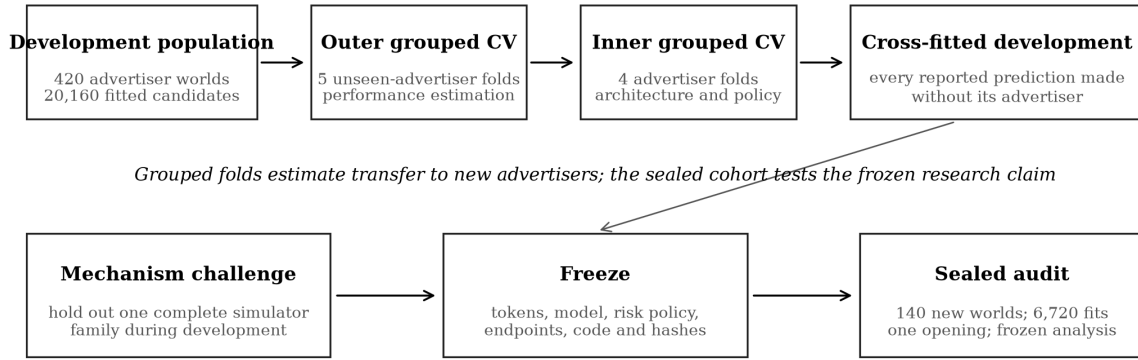


Figure 5. Validation governance. Nested cross-validation estimates advertiser transfer during development. The separately generated sealed cohort is opened exactly once after the method, posterior contract, endpoints, and source hashes are frozen.

10.2 One-time sealed audit

The confirmatory cohort contains 140 new advertiser worlds - 20 from each of seven mechanism families - and 6,720 new FullRankADVI candidate fits. Its seed window is disjoint from development. Before cohort generation, the audit contract froze the 232-token registry, selector, risk policy, source code, inference engine, bootstrap procedure, family composition, and endpoints. Technical failures could be rerun only under the identical payload and posterior contract. No score, token, threshold, or endpoint could be repaired after truth was opened.

10.3 Predeclared endpoints

Endpoint	Definition	Role
Mean excess loss	average loss above the in-pool oracle	central decision performance
P90 and P95	upper quantiles of selected excess loss	downside protection
Dangerous false champions	selected candidates above the declared regret threshold	unsafe top-one promotion
Oracle recall	fraction selecting the lowest-loss in-pool candidate	ranking sharpness
Risk-coverage area	loss integrated from 60% to 100% promotion coverage	quality of abstention
Theorem violations	selected capped regret above equation (6) receipt	implementation consistency

Table 5. Absolute confirmatory endpoints. No internal-version comparison is needed to interpret these quantities.

10.4 Inference and multiplicity

Family-stratified bootstrap resampling preserves the audit's equal family design. The primary comparison uses 10,000 resamples and a fixed seed. Point estimates are reported for every predeclared absolute endpoint, all seven mechanism families, paired business outcomes, and the complete risk-coverage curve. Analyses not named in the frozen protocol are descriptive and are not used to alter the confirmatory conclusion.

11 Internal sealed audit of RegretSet-MMM

The one-time internal audit evaluated RegretSet-MMM on 140 previously unseen advertisers generated by the study's synthetic data-generating process and 6,720 newly fitted candidate MMMs. This audit predates the evidence-adaptive extension in Section 8.4 and is retained unchanged as validation of the foundational set-wise method. All 6,720 candidate rows were finite, valid, and unique. The audit was scored once after posterior inference and truth-blind token construction were complete. All nine predeclared confirmatory checks passed.

Confirmatory quantity	100% coverage	70% coverage	Interpretation
Mean excess loss	1.10	0.79	average opportunity lost relative to the in-pool oracle
Median excess loss	0.42	0.26	typical selected-model loss
P90 excess loss	2.78	1.94	loss not exceeded by 90% of promoted advertisers
P95 excess loss	4.26	3.05	more extreme downside
Decision-risk objective	1.69	1.19	65% mean plus 35% P90 excess loss

Confirmatory quantity	100% coverage	70% coverage	Interpretation
Dangerous false champions	32.1%	27.6%	materially harmful top-one selections
Oracle recall	15.0%	18.4%	exact recovery of the lowest-loss candidate
Regret-bound violations	0	0	implementation check for equation (6)

Table 6. Absolute sealed-audit performance. Coverage is the fraction of advertisers for which the selector promotes a candidate rather than abstaining.

11.1 Primary confirmatory inference

The predeclared primary estimand was the full-coverage decision-risk objective for RegretSet-MMM minus the same objective for a frozen restricted-diagnostic baseline. The baseline used the same candidate set and economic objective but only a smaller predeclared subset of deployment-observable diagnostics. The observed difference was -1.456 loss units. The family-stratified two-sided 95% bootstrap interval was $[-2.837, -0.479]$, and the predeclared one-sided 95% upper bound was -0.601. All values favor RegretSet-MMM because lower loss is better. The interval excludes zero in the favorable direction, satisfying the superiority rule. Across individual businesses, RegretSet-MMM had lower loss in 56.4%, equal loss in 15.7%, and higher loss in 27.9%.

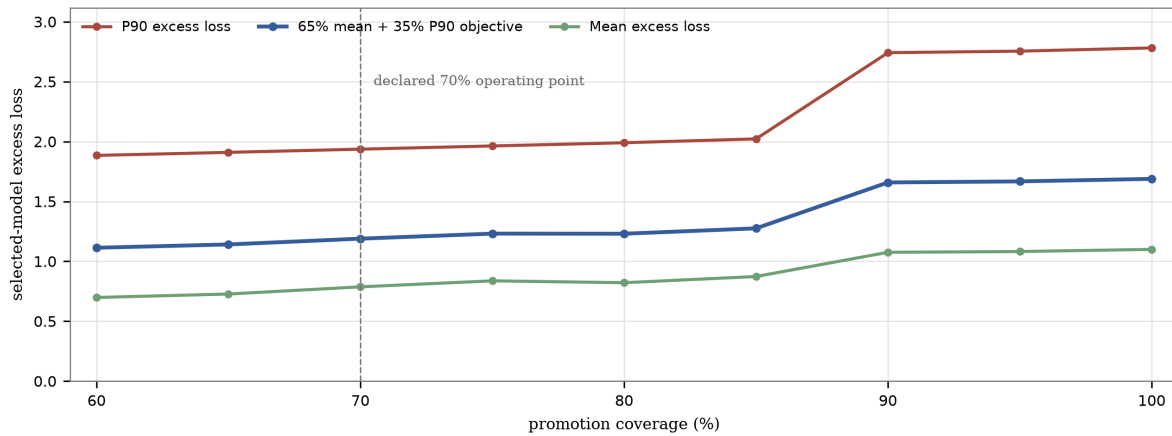


Figure 6. Absolute risk-coverage behavior on the sealed cohort. Abstaining on the least-confident advertisers reduces mean, tail, and composite decision loss. The 70% operating point was declared before the audit was opened.

11.2 Mechanism-family stress results

The overall result does not imply uniform difficulty. Social-frequency-pressure advertisers produced the largest mean and P90 losses; swapped-channel-mechanics produced the largest P95 loss despite a relatively low median. Delayed-TV advertisers had the highest dangerous-false-champion share. This heterogeneity supports reporting the full loss distribution rather than a single average.

Synthetic mechanism family	Mean	P90	P95	Danger / oracle
Balanced DTC	1.35	2.95	3.35	40% / 20%
Correlated planning	0.77	1.90	2.40	30% / 20%
Delayed TV	0.92	1.72	1.98	45% / 15%
Search demand harvesting	0.72	2.04	2.09	25% / 10%
Social frequency pressure	1.53	4.33	5.36	40% / 15%
Swapped channel mechanics	1.18	1.94	5.45	15% / 15%
Wrong external evidence	1.24	3.31	4.00	30% / 10%

Table 7. Full-coverage results by predeclared mechanism family, with 20 unseen advertisers per family. The final column reports dangerous-false-champion share and exact in-pool oracle recall.

11.3 Confirmatory checks and claim boundary

The primary superiority check, full-coverage mean and P90 improvements, P95 non-inferiority, dangerous-false-champion improvement, selective-risk-area improvement, matched 70% objective improvement, theorem verification, and FullRankADVI-only inference check all passed. The finite-candidate bound had zero violations. The result therefore confirms synthetic generalization under the frozen cohort contract. It does not establish real-world generalization, global optimality, or realized advertiser profit improvement. Any future selector tuning requires a new sealed cohort.

11.4 Operational provenance

All 6,720 posterior fits completed under the frozen inference contract. During local label assembly, the orchestration parser mapped zero-based shard 0 to shard 1; consequently, seven label shards completed while 18 businesses in shard 0 were never executed. The missing 864 candidate labels were generated locally with a temporary parser correction that affected shard indexing only. The scientific method, posterior artifacts, selector, endpoints, and hidden outcomes remained unchanged; hidden outcomes were not inspected before recovery. All eight shards were then hash-checked and assembled once before the frozen scorer ran once. The machine-readable recovery receipt independently records that the scientific method, selector, and posterior did not change.

12 External transport validation in AMSS

12.1 Fresh confirmatory test in a separately authored simulator

The Aggregate Marketing System Simulator (AMSS) was developed independently of FluxMMM to generate aggregate marketing time series and associated ground-truth return and marginal-return measures [23]. It therefore provides a stronger transport test than drawing additional seeds from the study's own generator. A new cohort, with seeds disjoint from the earlier AMSS study, was generated only after the evidence-adaptive architecture, 232 candidate tokens, 63 business-context tokens, FullRankADVI contract, candidate library, evidence rules, four decisions, loss definition, comparator family, and 65/35 mean-tail utility were frozen. No AMSS truth was available during fitting or selection, and no model was retrained or reselected after truth opening.

The external cohort contains 100 AMSS businesses, balanced across paid-social-experiment, search-experiment, television-experiment, and no-experiment evidence conditions. Each business contributes 24 structural specifications under two evidence arms, producing 4,800 FullRankADVI fits. Each posterior fixed actions for budget reduction, fixed-budget reallocation, budget growth, and an economic-ceiling decision before truth was opened, for 19,200 evaluated actions. The unit of evaluation is the business.

The design uses a deterministic 100-point marginal space-filling cohort and rotates all six orders of the three media modules. Candidate fitting observes 156 weekly periods after a 52-week burn-in. Frozen actions cover a 52-week decision horizon plus eight post-action carryover weeks. Every candidate chooses the feasible allocation that maximizes posterior-expected incremental contribution less incremental media cost. AMSS then evaluates the fixed action using four common-random-number truth replicates. Period-matched experiments, when assigned, compare treatment with a 20%-lower-budget path from the same pretest state and include a declared post-test outcome window.

12.2 Selection rules and benchmark hierarchy

Evidence-adaptive RegretSet-MMM. The focal learned rule. It retains RegretSet-MMM's joint candidate-set representation and adds the advertiser-level evidence context and residual gate described in Section 8.4. It sees only truth-blind deployment tokens.

RegretSet-MMM. The frozen non-adaptive learned rule. It uses the same 232 candidate tokens, economic-loss targets, and joint candidate-set pathway but no 63-token advertiser evidence context or gated residual experts.

Prediction-only. Selects the valid candidate with the strongest frozen three-fold rolling out-of-sample score. It is a strong deployable baseline because it uses no hidden economic truth.

Classic Pareto. Forms a nondominated set over rolling prediction error, DECOMP.RSSD, and same-window experiment MAPE when available, then applies a deterministic equal-rank compromise. It is an additional benchmark rather than a member of the predeclared primary screen.

Uniform random and in-pool oracle. Uniform random averages over all 48 valid choices within each business. The oracle observes hidden AMSS truth and supplies an unattainable lower reference within the same candidate pool; neither is a deployable selection rule.

Selection rule	Status	Mean	P90	P95	Objective
Evidence-adaptive RegretSet-MMM	confirmatory target	0.135	0.329	0.401	0.203
Prediction-only	primary comparator	0.217	0.605	0.909	0.353
RegretSet-MMM	primary comparator	0.247	0.688	0.907	0.401
Uniform random valid	descriptive chance	0.395	0.576	0.616	0.458
Classic Pareto	additional benchmark	0.539	1.000	1.000	0.701
In-pool oracle	truth-informed lower bound	0.017	0.052	0.064	0.029

Table 8. Fresh AMSS capped decision risk. The objective is 65% cohort mean plus 35% cohort P90; lower is better. Uniform-random rows summarize each business's conditional expected risk. Classic Pareto is an additional, non-confirmatory benchmark. The oracle is not deployable.

12.3 Confirmatory result

Evidence-adaptive RegretSet-MMM achieved a decision-risk objective of 0.203. RegretSet-MMM achieved 0.401; the paired evidence-adaptive-minus-RegretSet-MMM difference was -0.199, with 95% bootstrap interval [-0.292, -0.080]. Prediction-only selection achieved 0.353; the paired difference was -0.151, with interval [-0.298, -0.054]. Thus the evidence-adaptive selector reduced the objective by 49.5% relative to RegretSet-MMM and 42.6% relative to prediction-only selection.

All three predeclared checks passed: the evidence-adaptive objective was lower than RegretSet-MMM, no higher than prediction-only selection, and its P95 capped risk of 0.401 remained well inside the 0.02 non-inferiority margin relative to the better comparator tail. Both paired objective intervals exclude zero in the favorable direction. The bootstrap resampled businesses within the four evidence groups for 10,000 replicates, preserving the design's evidence balance.

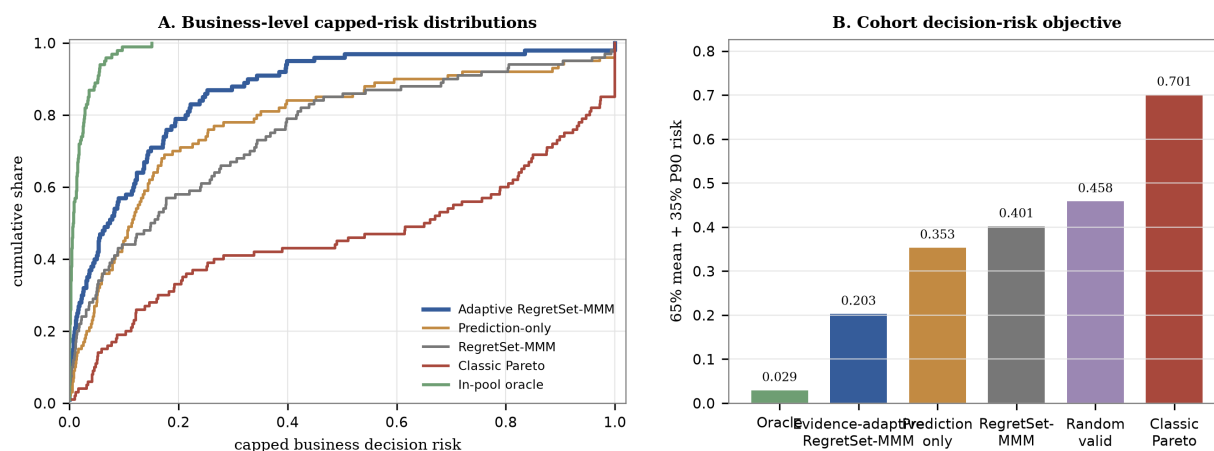


Figure 7. Fresh AMSS transport. Panel A shows business-level capped-risk distributions for deployable selectors and the in-pool oracle. Panel B reports the common 65/35 mean-tail objective. Classic Pareto is additional rather than confirmatory.

12.4 Additional classic Pareto benchmark

Classic Pareto approximates established multi-objective MMM practice [7]. It forms the nondominated set over rolling prediction error, DECOMP.RSSD between analytic-screening spend and contribution shares, and same-window experiment MAPE when an experiment exists. A deterministic equal-rank compromise selects one member; no-experiment businesses use prediction and decomposition only.

Classic Pareto's objective was 0.701. Adaptive RegretSet-MMM was 71.1% lower. Its business-level mean-risk difference was -0.405, with 95% interval [-0.469, -0.337]; adaptive RegretSet had lower risk for 88 businesses, tied for 2, and higher risk for 10. This supports superiority to the stated Pareto rule in this cohort, not to every Pareto implementation, expert review process, or established MMM system.

12.5 Interpretation and remaining headroom

The descriptive uniform-random objective was 0.458, 55.8% above adaptive RegretSet. The evidence-group-stratified paired mean-risk difference against each business's random-valid expectation was -0.260, with 95% interval [-0.287, -0.232]. The in-pool oracle achieved an objective of 0.029. The gap from 0.203 demonstrates substantial headroom even without expanding the candidate library. The result supports external transport of the learned ranking rule, not global optimality.

The channel adapter maps AMSS native search to a nonbrand-search archetype, traditional television to the long-memory CTV role, and an unchanged AMSS traditional-media module to a paid-social stress proxy. The proxy preserves AMSS delivery, reach, frequency, state-transition, and stochastic mechanisms but is not a literal platform auction. This evaluation therefore tests transport across independently implemented marketing mechanisms and endogenous delivery, not exact platform-label fidelity.

All 100 truth partitions and 19,200 unique candidate-scenario outcomes were retrieved and hash-verified. The protocol forbade post-truth retraining, reselection, and business exclusion. These controls support a confirmatory external synthetic transport claim for the two predeclared comparators; they do not establish realized effects in advertiser data or universal superiority to every published MMM workflow.

13 Interpretation for researchers and practitioners

13.1 What the selector learns

The learner estimates a conditional mapping from an advertiser's joint candidate evidence to economic risk. It does not learn universal causal coefficients, and its hidden embedding is not an advertiser identity. At deployment it asks: among these candidates, which posterior and diagnostic configuration resembles configurations that led to low decision loss in held-out synthetic advertisers?

13.2 RegretSet-MMM development evidence: what predicted lower economic loss

Raw neural-network weights are not stable scientific importance measures because nonlinear encoders distribute information across correlated tokens. The following pre-adaptation analysis concerns RegretSet-MMM and isolates what its joint 232-token pathway learned. We organize the tokens into six exhaustive practitioner pillars. For each pillar g , the entire group is removed, RegretSet-MMM is relearned on the outer-training advertisers under the originally selected architecture and policy, and economic loss is measured on the untouched outer fold. The primary importance estimand is

$$I_g = 100 \frac{J_{-g}^{OOF} - J_{all}^{OOF}}{J_{all}^{OOF}}, \quad (21)$$

where J is the declared 65% mean / 35% P90 selected-model excess-loss objective. Positive I_g means the remaining tokens could not replace the removed information; negative I_g means the group was conditionally redundant or harmful at the available sample size. The sign is not a causal statement about the diagnostic and does not license removing a scientific safety check. A twenty-repeat within-advertiser permutation analysis is retained as a sensitivity check, but drop-column refitting is primary because it permits the learner to adapt to the information that remains.

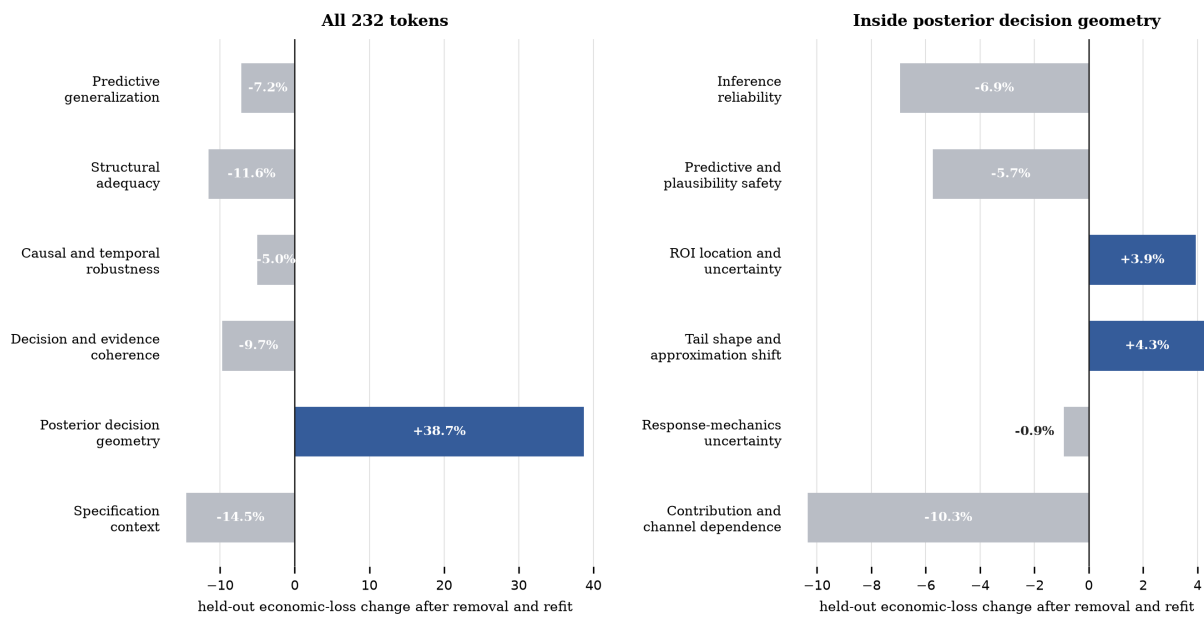


Figure 8. RegretSet-MMM advertiser-grouped development importance. Bars show the percentage change in held-out economic loss when a complete token group is removed and RegretSet-MMM is relearned. Positive values indicate incremental predictive information; gray negative values indicate conditional redundancy or finite-sample overfit, not that the underlying scientific check is unnecessary. The sealed cohort was not opened for this analysis.

The central result is unusually concentrated. Removing posterior decision geometry increased held-out economic loss by 38.7%, and the direction was adverse in four of five advertiser folds. Removing any other broad pillar reduced the aggregate objective: causal and temporal robustness -5.0% , predictive generalization -7.2% , decision and evidence coherence -9.7% , structural adequacy -11.6% , and specification context -14.5% . The permutation sensitivity analysis also disrupted posterior geometry in all 20 repetitions; specification context appeared important under permutation but not after refitting, a pattern consistent with interaction information that is used by the fitted network yet can be relearned from correlated posterior summaries.

A second ablation partitions the 105 posterior-geometry tokens. Removing tail shape and approximation-shift summaries increased held-out loss by 4.3% and did so in four of five folds; removing ROI location and uncertainty increased loss by 3.9% in three of five folds. The other posterior subgroups were replaceable after refitting. These subgroup effects do not add to the 38.7% whole-pillar effect: the network is nonlinear, and the full bundle carries interactions and jointly redundant signals that are destroyed together but recoverable when only one subgroup is absent.

Within this synthetic development population, the actionable lesson is therefore not that classical diagnostics cease to matter. It is that the strongest incremental signal for economic transfer came from the posterior state presented to the decision maker: the location and width of channel ROI distributions, implausible and near-zero tail mass, asymmetry and approximation shift, contribution stability, response-parameter uncertainty, and cross-channel covariance. Point fit scores or a named adstock family were not sufficient on their own. A candidate was more likely to transfer economically when its full posterior described a stable and decision-usable allocation problem.

Practitioner pillar	Tokens represented	Development finding and use
Posterior decision geometry	ROI levels and intervals; tail mass; contribution stability; response uncertainty; channel covariance; FullRankADVI reliability	+38.7%. Inspect the entire posterior allocation geometry before promoting a model.
Model specification context	adstock, saturation, dynamics, likelihood, calibration and their declared interactions	-14.5%. A named specification is context, not a transferable quality score by itself.
Causal and temporal robustness	placebos, anchor recovery, confounder stress, whole-flight and carryover support	-5.0%. Preserve as causal challenge evidence; current tokens added no robust incremental ranking information.
Predictive generalization	rolling forecasts, coverage, fold stability and high/low-spend regimes	-7.2%. Keep as transparent validation evidence; avoid equating forecast skill with decision quality.
Structural adequacy	residual, variance, likelihood-shape, influence, rank and boundary checks	-11.6%. Use for model criticism and failure diagnosis, not as a stand-alone promotion score.

Practitioner pillar	Tokens represented	Development finding and use
Decision and evidence coherence	ROI plausibility, source quality and conflict, identification and evidence-deletion decisions	−9.7%. Retain source-specific receipts; simplify correlated inputs before retraining the selector.

Table 10. Practitioner interpretation of the RegretSet-MMM outer-fold development ablation. Percentages are conditional predictive importance within the study's synthetic development data, not evidence-adaptive routing weights, universal validation weights, or confirmatory audit outcomes.

13.3 A disciplined interpretation of the null pillars

A negative drop-column result does not imply that residual tests, temporal placebos, experiments, or business plausibility are dispensable. First, several appear again inside posterior geometry: weak identification expands intervals, collinearity induces channel covariance, and conflicting evidence moves posterior location or tail mass. Second, a diagnostic can be essential as a hard safety gate even when it adds no marginal ranking accuracy. Third, correlated token groups divide predictive credit. The result instead identifies a model-development priority: compress overlapping diagnostics, preserve raw validation receipts for human review, and require future selector versions to demonstrate positive outer-fold ablation before claiming a pillar improves economic ranking.

13.4 What happens for a new advertiser

The system validates data cadence and schema, generates a declared candidate set, fits FullRankADVI posterior approximations, computes the 232 truth-blind tokens, and applies the frozen set-wise selector. It reports predicted mean and tail loss, confidence, scenario risk, ROI and contribution uncertainty, source attribution, source conflict, and source-deletion decision dependence. Low-confidence cases are reviewed or abstained. The final production posterior may be refitted with a higher-fidelity sampler; any posterior shift is disclosed rather than silently overwriting the validation receipt.

13.5 Why plausible ROI is necessary but not sufficient

An industry-compatible average ROI can coexist with the wrong carryover, saturation, or channel covariance and therefore generate a poor marginal allocation. Conversely, a posterior may conflict with a broad benchmark because the advertiser genuinely differs. RegretSet-MMM uses plausibility, evidence quality, conflict, identification, and decision dependence as separate tokens. It never repairs a fitted ROI after seeing the economic label.

14 Limitations

- Synthetic transport: AMSS is independently authored and materially strengthens external validity, but both development and external evaluation remain simulated rather than realized advertiser interventions.
- Candidate-pool dependence: the oracle is best among 48 fitted candidates under one decision contract, not the true global model. A stronger search can change both labels and selector behavior.
- Approximate posterior inference: FullRankADVI may distort skew, multimodality, and tail dependence. All reported candidate posteriors use the same FullRankADVI contract, so the conclusions are conditional on that approximation.
- Restricted channel universe: three archetypal channels cannot represent the full granularity, geography, funnel stages, retail media, creative, or measurement systems of large advertisers.
- Simplified experiments: external evidence is period-matched and quality-annotated but does not reproduce every auction, compliance, interference, platform-measurement, or geo-spillover mechanism.
- Economic utility: conversion from outcome to contribution, allocation constraints, four scenario weights, and the 65/35 mean-tail preference are business choices; different utilities can produce different rankings.
- Evidence attribution is local: curvature shares describe sensitivity near one posterior approximation, not global causal validity or a Shapley decomposition of all modeling choices.
- Neural-selector uncertainty: bootstrap disagreement is an operational proxy rather than a full Bayesian posterior over the meta-model.
- No real-world causal endpoint: the current design cannot establish that the learned rule improves realized advertiser profit.

14.1 Interpretive boundaries

The completed internal and AMSS audits establish transfer only under their declared synthetic populations, candidate pool, evidence rules, and decision utility. They do not establish universal state-of-the-art MMM performance, global optimality, superiority to expert modelers, or realized external validity across industries. Nor do they imply that evidence influence proves evidence quality, residual diagnostics prove exogeneity, or the finite-candidate regret bound proves population generalization.

15 Open science and reproducibility

FluxMMM is the open-source implementation of RegretSet-MMM. Scientific artifacts are content-addressed: simulator contract, candidate fingerprints, posterior settings, token registry, grouped folds, selector configuration, source hashes, and audit endpoints are versioned. Truth-leakage tests prohibit hidden labels from candidate fitting and deployment tokens. Permutation tests verify candidate- and channel-order invariance. Numerical tests verify the finite-candidate bound and evidence-share exhaustiveness.

Stage	Permitted information	Frozen receipt
World generation	observed history and hidden truth	seed, generator family, parameters, decision contract
Candidate fitting	observed data and declared evidence	candidate and FullRankADVI fingerprint
Token construction	fit, diagnostics, metadata allowed by registry	232-token names, hashes, leakage and permutation tests
Decision labeling	hidden truth only after action	four actions, oracle actions, absolute and excess losses
Selector training	development labels only	nested folds, hyperparameters, ensemble seeds
Internal sealed audit	truth opened once after freeze	all endpoints, intervals, failures, provenance
External AMSS audit	frozen actions; AMSS truth opened once	adapter contract, actions, hashes, conditional random benchmark, verified truth receipt
Deployment	no causal truth	predicted risk, confidence, evidence profile, candidate fingerprint

Table 11. Information and artifact contract.

Open sourcing allows researchers to challenge the simulator, contribute new channel mechanisms, add decision utilities, reproduce posterior tokens with different inference engines, and submit selectors under the same truth firewall. Revisions after an opened audit must receive a new method and audit identifier. Negative results are part of the public scientific record.

15.1 Code and data availability

The canonical project website is <https://fluxmmm-web.web.app/>. Source code, model contracts, tests, and reproducibility materials are available in the FluxMMM repository at <https://github.com/gustavobramao/fluxmmm>. The public distribution identifies which artifacts are executable source, cached posterior output, or externally generated simulation evidence.

16 Conclusion

RegretSet-MMM reframes the selection of a marketing mix model as a learned, cross-advertiser decision problem. Candidate models are fitted without causal truth, summarized by named posterior and validation tokens, compared jointly as an unordered set, and supervised by the economic consequences of their budget actions. The evidence-adaptive version also conditions residual routing on the advertiser's evidence environment while retaining the complete joint token path. The method predicts the distribution—not only the mean—of loss and can abstain when top-one choice is ambiguous.

Two theoretical components make the design auditable. The finite-candidate inequality shows why economically costly oracle misrankings bound selected capped regret and motivates gap-weighted ranking. The source decomposition reveals whether local ROI precision is supplied by observational data, experiments, benchmarks, or generic regularization, without confusing influence with validity.

The internal sealed audit supports RegretSet-MMM on previously unseen advertisers generated by the study's synthetic data-generating process. More importantly, a fresh AMSS cohort confirmed that evidence-adaptive RegretSet-MMM lowered the 65/35 mean-tail decision-risk objective relative to RegretSet-MMM and prediction-only selection, with favorable paired bootstrap intervals and preserved P95 tail safety. Classic Pareto and uniform random valid selection produced higher loss under their stated rules. Because no fresh-cohort truth was available during selector training, candidate fitting, or action selection, the result demonstrates that deployment-observable posterior and validation evidence can carry useful economic-ranking information across synthetic data-generating systems.

The sizeable gap to the in-pool oracle, the restricted channel archetypes, the FullRankADVI approximation, and the absence of real-advertiser causal outcomes define the next research frontier. RegretSet-MMM should therefore be read as an externally transportable and falsifiable decision-focused selection protocol, not a claim of universal model recovery. Its open implementation allows researchers and advertisers to reproduce the evidence, test new simulators and candidate libraries, and challenge the learned rule under new decision utilities.

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A Appendix: frozen research contract

A.1 Study dimensions

Object	Frozen value
Development advertisers	420
Candidates per advertiser	48
Development candidate fits	20,160
Outer / inner grouped folds	5 / 4
Posterior engine	PyMC 6.2 FullRankADVI
Tokens per candidate	232 (136 base + 96 posterior decision)
Selector	joint-plus-evidence-gated DeepSets, 10 outputs
Business evidence context	63 candidate-independent tokens with missingness masks
Bootstrap ensemble	5 fits
Sealed audit	140 advertisers $\times$ 48 candidates = 6,720 new fits
Sealed families	7, with 20 advertisers per family
External AMSS audit	100 businesses $\times$ 48 candidates = 4,800 new fits
External frozen actions	19,200 across four budget decisions

Table A.1. Core study dimensions.

A.2 Architecture and policy grid

Choice	Values considered inside inner grouped folds
Joint / expert encoder width	12 or 20
Decoder width	12 or 16
Epochs	18 or 22
Learning rate	0.0022 or 0.0018
Weight decay	0.001
Set-selection temperature	0.35
Danger penalty	0, 0.25, or 0.50
Ensemble disagreement penalty	0.25
Mean / P90 risk preference	0.65 / 0.35

Table A.2. Evidence-adaptive development choices. Outer-fold outcomes do not select these values.

B Appendix: posterior decision token registry

B.1 Twenty-three channel metrics

Group	Metrics before four symmetric aggregations
ROI	log median; log mean; log lower; log upper; log relative width; log draw SD
Safety and shape	implausible probability; near-zero probability; skewness; location shift; interval overlap
Evidence	experiment indicator; benchmark indicator
Adstock	mean; SD; log Weibull scale mean; Weibull shape mean; Weibull share; sum-normalization share
Saturation	mean; SD; half-saturation mean
Contribution	log coefficient of variation

Table B.1. Channel metrics. Mean, SD, minimum, and maximum over channels produce 92 tokens.

B.2 Four pairwise posterior summaries

- Mean absolute pairwise correlation of channel log-ROI draws.
- Maximum absolute pairwise correlation of channel log-ROI draws.
- Mean absolute pairwise correlation of channel contribution draws.
- Maximum absolute pairwise correlation of channel contribution draws.

C Appendix: algorithmic protocol

C.1 Development

1. Generate advertiser worlds and seal hidden response truth.
2. Fit the common 48-candidate library with the immutable FullRankADVI contract.
3. Compute 232 truth-blind tokens and verify candidate- and channel-order invariance.
4. Optimize four actions from each candidate posterior under common feasibility constraints.
5. Reveal truth only to label action loss, in-pool oracle loss, excess loss, and danger.
6. Construct matched full, experiments-only, and benchmark-gap-fill candidate sets; keep all sets from one advertiser in the same fold.
7. Use nested advertiser-grouped cross-validation to select architecture and policy and obtain outer-fold predictions.
8. Freeze the evidence-adaptive method and endpoints, then generate a fresh, disjoint AMSS cohort.

C.2 Deployment

1. Validate schema, cadence, evidence, and decision utility for a new advertiser.
2. Generate and fit the declared candidate set without causal truth.
3. Compute the identical 232 candidate tokens, 63 business-context tokens, and continuous evidence reports.
4. Apply the frozen evidence-adaptive set-wise ensemble to predict loss distribution and adjusted risk.
5. Promote, abstain, or request review under the declared coverage and risk policy.

D Appendix: sealed-audit provenance receipt

Audit item	Frozen completion value
Audit identifier	003; one-time confirmatory synthetic audit
Completion timestamp	2026-09-04T22:07:12Z
Successful / failed candidate fits	6,720 / 0
Advertisers / families	140 / 7; 20 advertisers per family
Result artifact SHA-256	dbfa626ce8caffb0d42afc86bd75930b4cd21a04010924c0843c9e82bb5f8159
Dataset SHA-256	0b34c975474e592c3c5cc7b4950ce744c59ec856511d6f02c1054d81ffd0fe5c
Protocol freeze SHA-256	1f23f8f963992265af2ab4d46005996b25b4c694f5b1eaaef4bd119d3e622c3
Primary difference	-1.456; two-sided 95% interval [-2.837, -0.479]
Confirmatory checks	9 of 9 passed
Theorem-bound verification	0 violations across 140 selected candidates
Inference contract	PyMC 6.2 FullRankADVI; NUTS excluded
Operational deviation	zero-based shard 0 was skipped by an argument-parser clamp; 18 missing businesses were recovered locally under an indexing-only patch before one-time scoring
Scientific changes during recovery	none: posterior, selector, tokens, endpoints, and hidden outcomes remained unchanged

Table D.1. Immutable completion receipt for the sealed audit. Full machine-readable provenance is distributed with the research artifacts.

E Appendix: external AMSS transport receipt

E.1 Frozen external contract

External-audit item	Frozen or verified value
Protocol	fresh AMSS confirmatory audit; frozen before truth
Simulator	AMSS 1.0.1; separately authored aggregate marketing system
Businesses / candidate fits	100 / 4,800
Decision actions	19,200; four per candidate
Evidence allocation	25 businesses each: paid-social, search, television, or no experiment
Visible / future horizon	156 / 52 weeks; 8 carryover weeks
Frozen selector	Evidence-adaptive RegretSet-MMM; 232 candidate + 63 context tokens
Posterior	PyMC 6.2 FullRankADVI; 5,000 iterations; 256 draws; two primary seeds
Primary loss	scenario excess loss capped at 1; business risk = 65% mean + 35% P90
Evaluated selection rules	evidence-adaptive RegretSet-MMM; RegretSet-MMM; prediction-only; uniform random valid; classic Pareto; in-pool oracle reference
Truth verification	100 partitions; 19,200 unique outcomes
Confirmatory checks	evidence-adaptive RegretSet-MMM vs RegretSet-MMM; evidence-adaptive RegretSet-MMM vs prediction-only; P95 tail safety - all passed
Evidence-adaptive development SHA-256	d01547789bd397329445bcd42e623ef06161fddc7df6e103014aa3c98b8b861f
Result SHA-256	928bffbe61987f91ed3ab38bb068c2845c4b8ed22a410dc5ba6ca86640e1f0b1
Candidate-loss SHA-256	37a34711e9a4970f2e733241a6e5add3e17686c1991f7419bd861c12a7e0cfd8

Table E.1. External AMSS transport contract and reproducibility receipt.

E.2 Random-valid aggregation and paired uncertainty

Every external business has 48 valid candidates. Uniform random valid selection is evaluated without Monte Carlo error by assigning probability $1/48$ to each candidate within a business. The resulting conditional expected risk is then summarized across the 100 businesses. Its mean, P90, P95, and 65/35 objective are 0.395, 0.576, 0.616, and 0.458. These are quantiles of business-level conditional expectations, not the tail of one realized random candidate draw.

The descriptive paired interval resamples businesses within each of the four 25-business evidence groups. With 10,000 resamples and seed 10031772, adaptive RegretSet-MMM minus expected-random mean risk was -0.260, with 95% interval [-0.287, -0.232]. Random-valid was not one of the two primary confirmatory comparators.

E.3 Channel-adaptor boundary

AMSS native search is mapped to the frozen nonbrand-search role and AMSS traditional television to the long-memory CTV role. The paid-social stress proxy is generated with AMSS's unchanged DefaultTraditionalMediaModule and mapped to the frozen paid-social role. Raw proxy naming and module provenance are preserved. The adaptor makes no claim that AMSS implements a native social-platform auction or that traditional television is literally connected television.

E.4 Additional classic Pareto receipt

Secondary item	Verified value
Status	additional benchmark outside the confirmatory two-comparator screen
Candidate pool	100 businesses $\times$ 48 unchanged valid candidates
Classic Pareto	rolling OOS; MAP DECOMP.RSSD; same-window experiment MAPE
Pareto compromise	lowest equal-weight average objective rank on the nondominated set
No-experiment rule	prediction and decomposition objectives only
New posterior fits / cloud compute	0 / none
Changed candidate actions	none; accepted sealed losses were joined by candidate ID
Artifact SHA-256	ecd9b42b7b0da916c017d4d6d61544cf413b234f99cf167eee86e4a462b78a3

Table E.2. Reproducibility and governance receipt for the additional classic Pareto benchmark.